Name: Project Week 10 (Bank Marketing Campaign)
Week 10: Deliverables
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Specialization: Data Science
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(Individual project)

Table of Contents

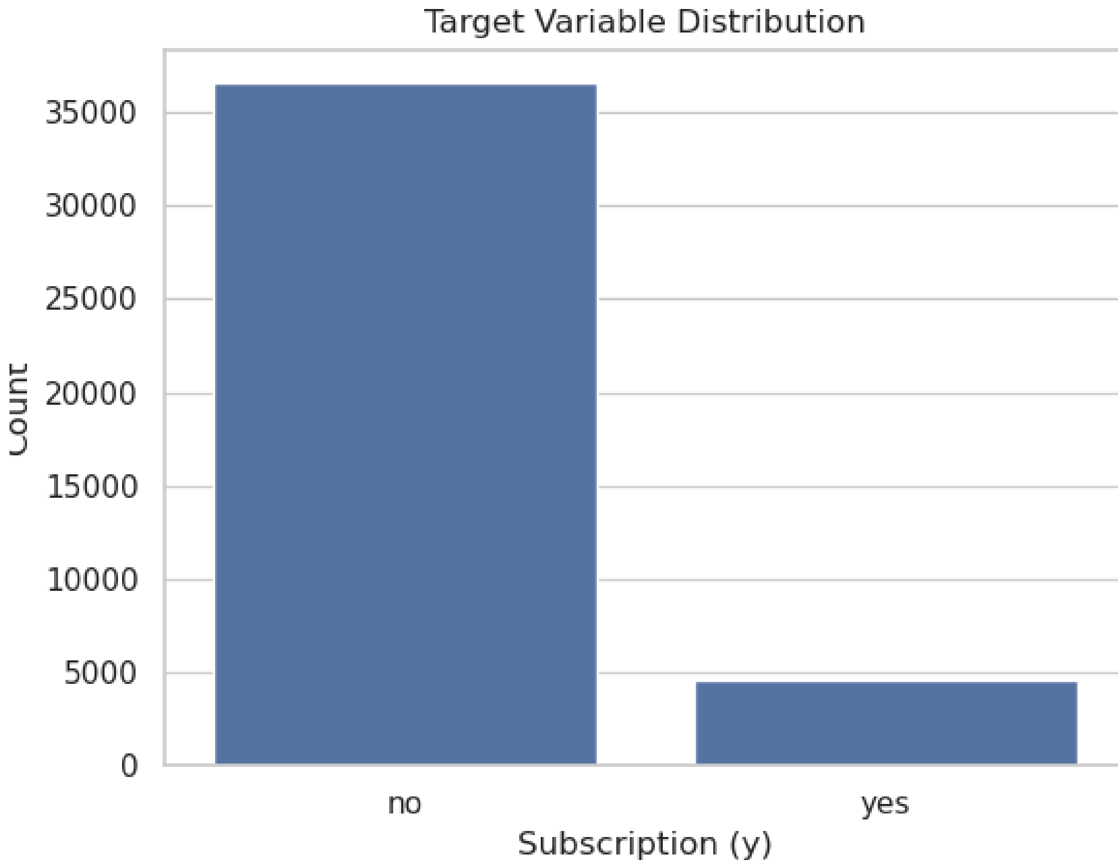
Solve the problems :

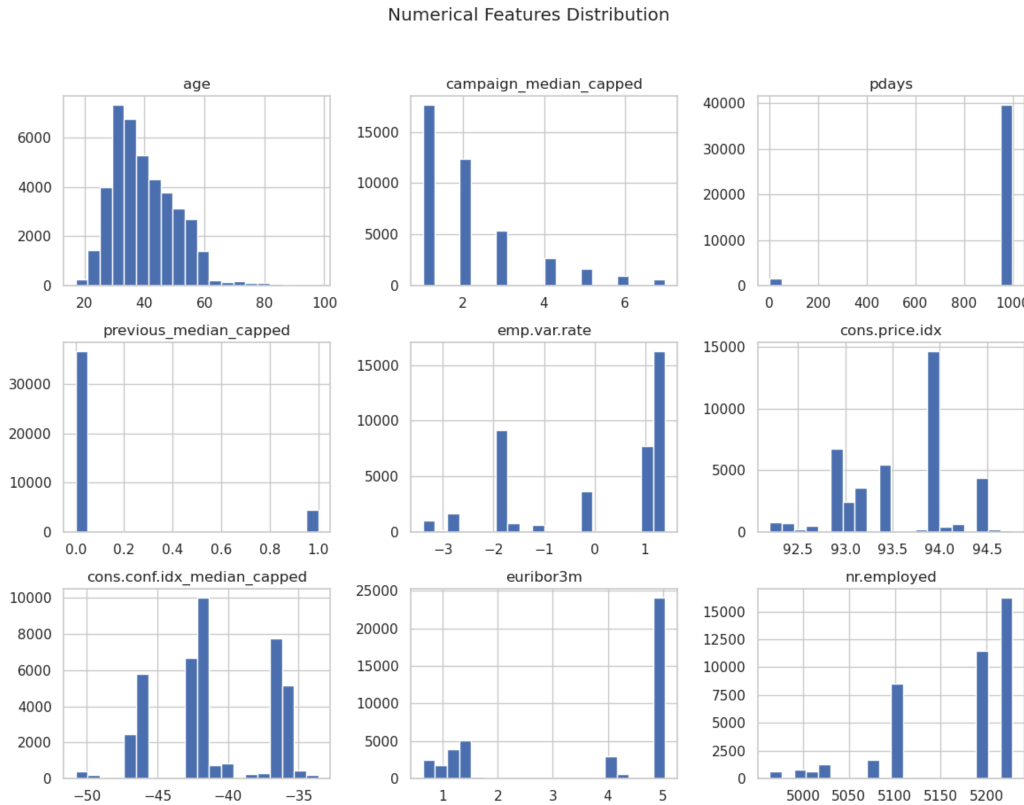
1. Problem Description

ABC Bank wants to sell it's term deposit product to customers and before launching the product they want to develop a model which help them in understanding whether a particular customer will buy their product or not (based on customer's past interaction with bank or other Financial Institution).

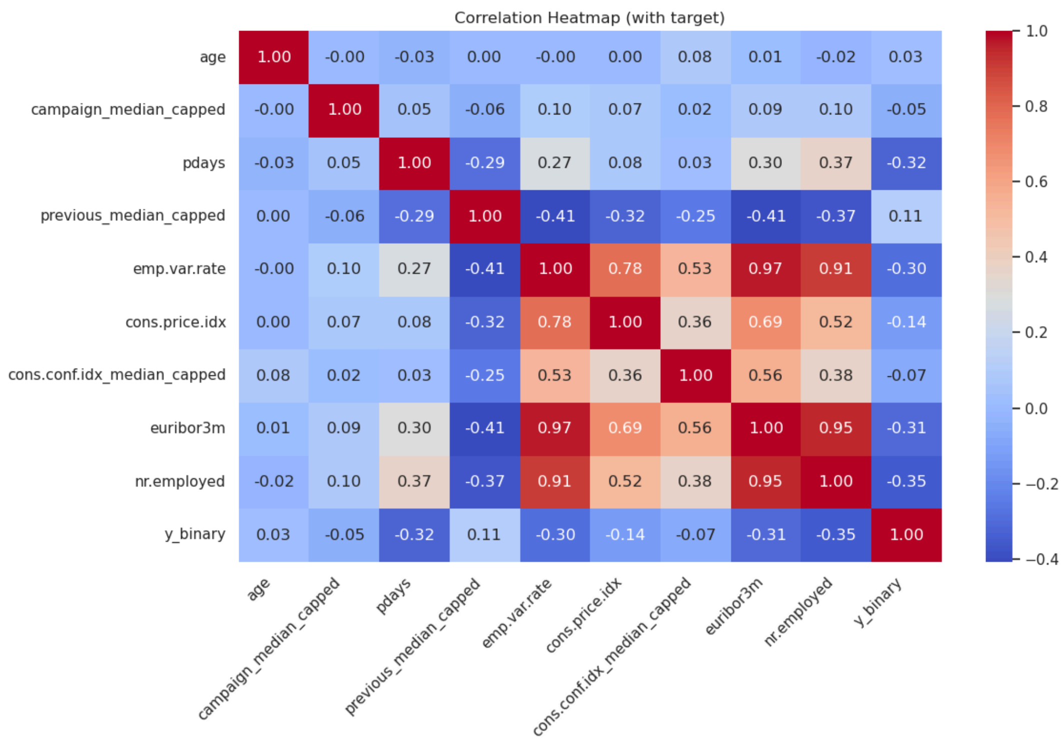
2. EDA performed on the data

- Exploring Numerical values

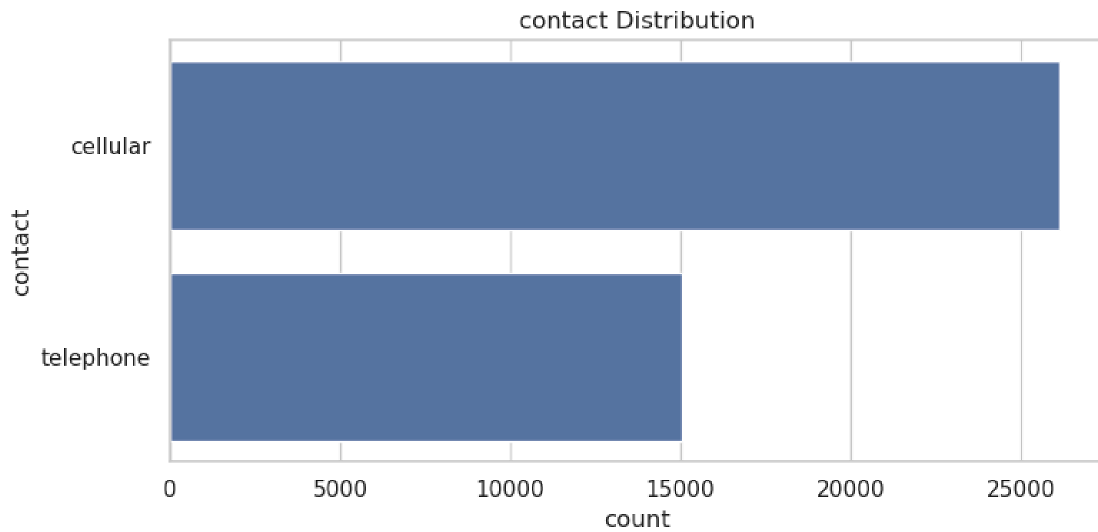

Target Variable Distribution
Count
35000
30000
25000
20000
15000
10000
5000
0
no
yes
Subscription (y)



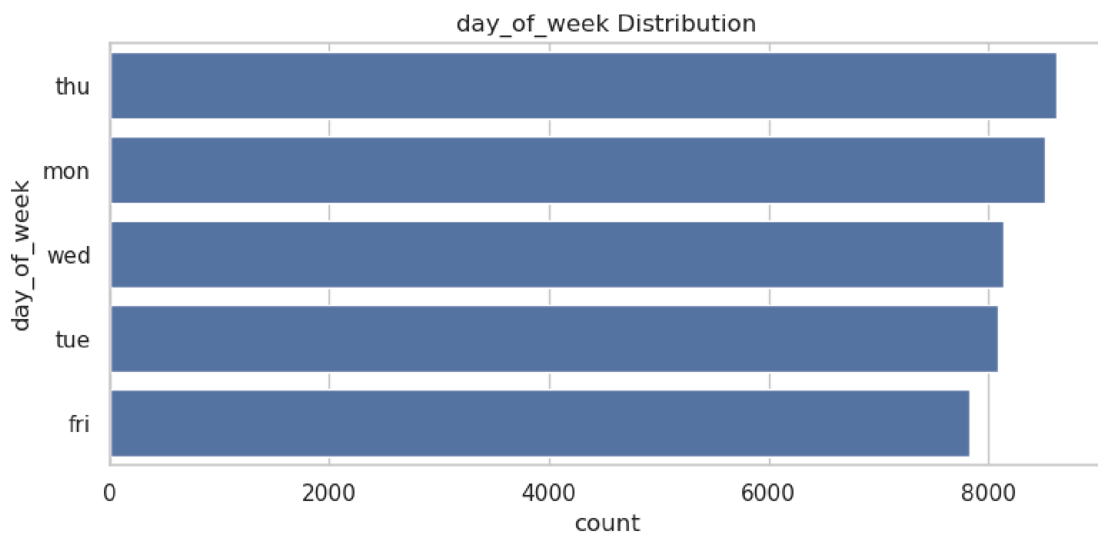
- Correlations among numerical feature along with the Output variable (desired target): y



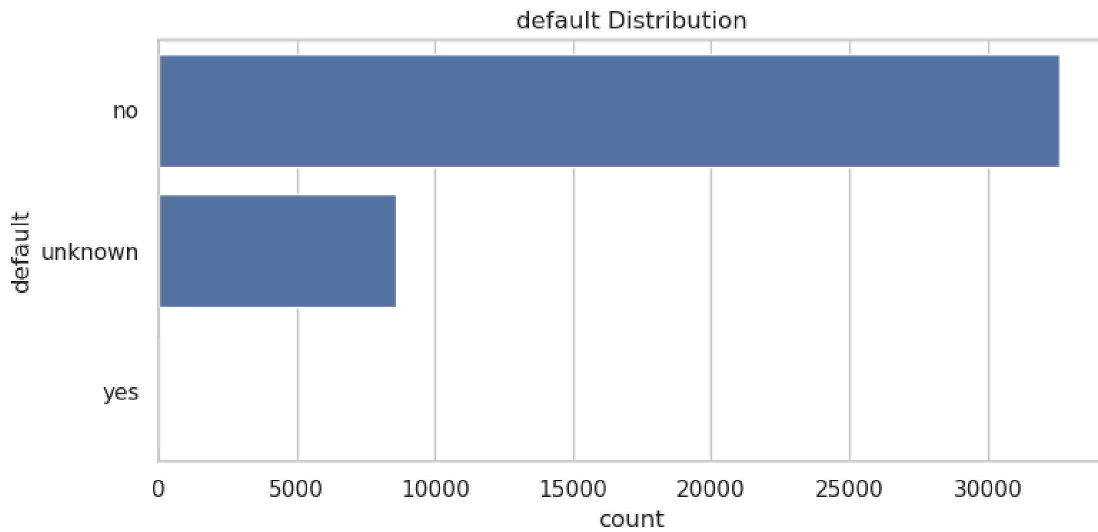
-
- Feature-wise Explanation
  - emp.var.rate (Employment Variation Rate): More employment variation → less likely to subscribe (economic uncertainty?)
  - euribor3m (Euro Interbank Offered Rate - 3 Months) :Higher interest rates → less subscriptions (people may avoid locking into term deposits)
  - nr.employed (Number of Employees): High employment → possibly lower urgency to invest or switch products
  - pdays: If recently contacted → more likely to subscribe, 999 = not contacted → less likely → explains negative correlation
- Exploring Categorical values

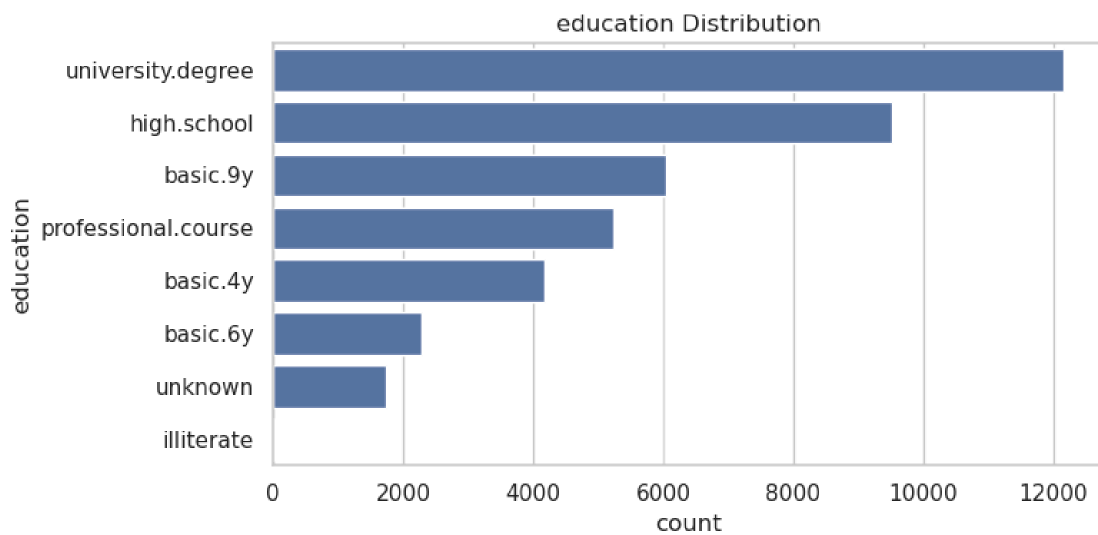


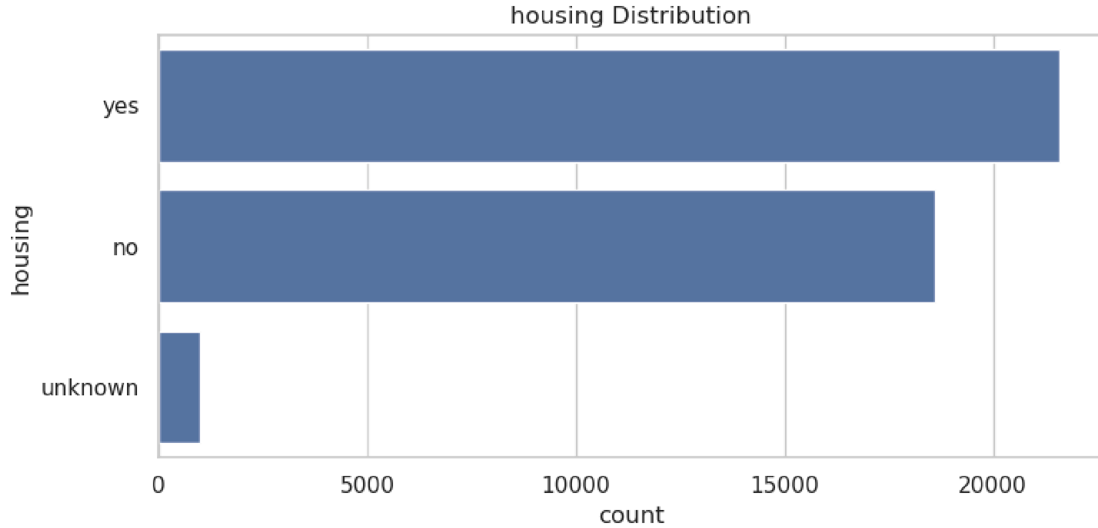
contact Distribution
cellular
telephone
contact
0
5000
10000
15000
20000
25000
count

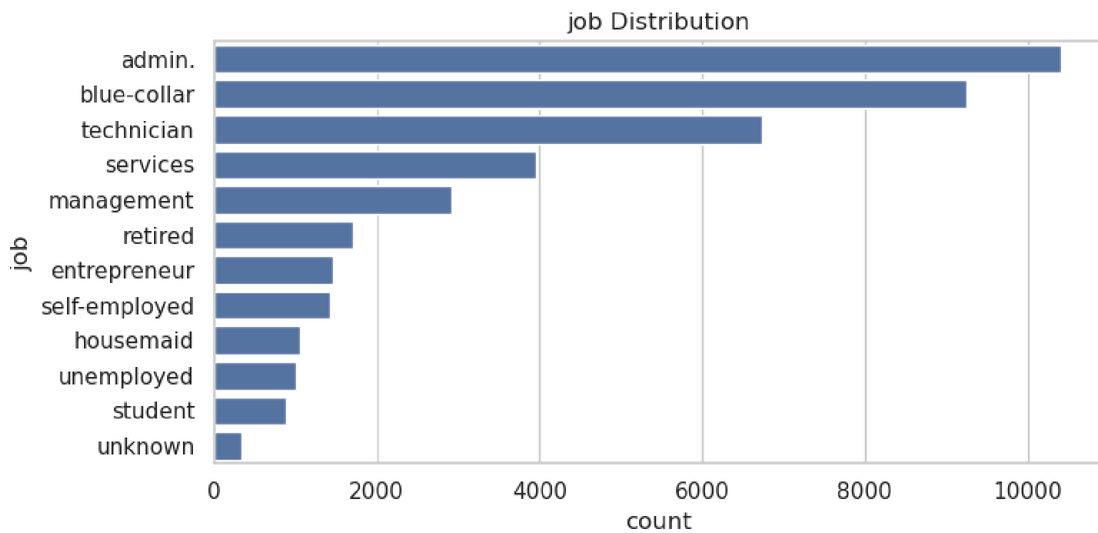


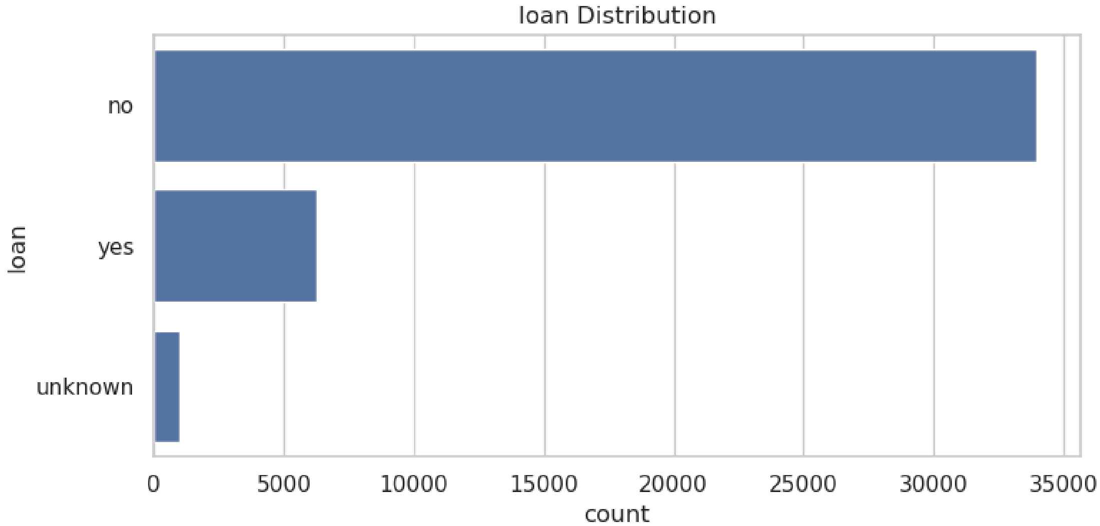
day_of_week Distribution
thu
mon
wed
tue
fri
day_of_week
0
2000
4000
6000
8000
count

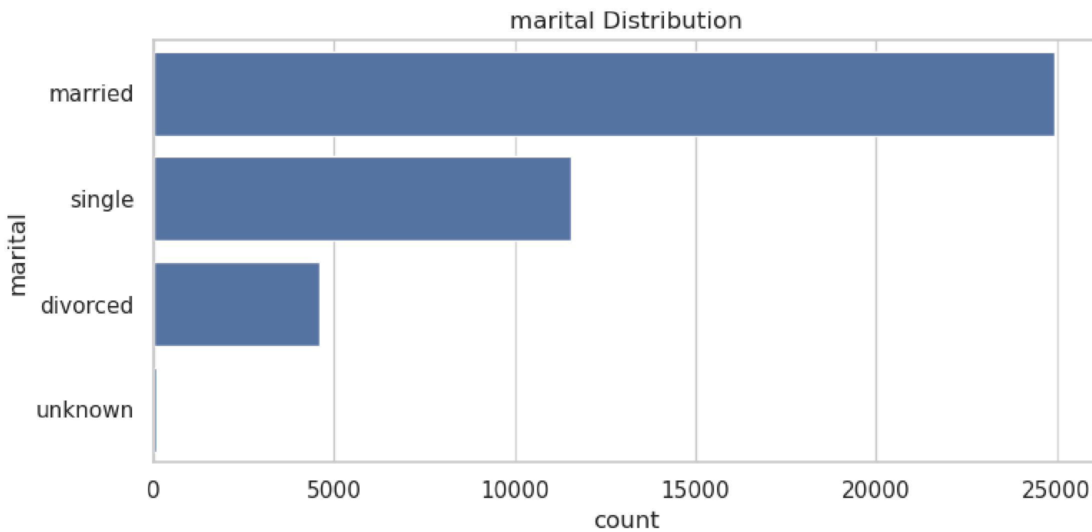

default Distribution
no
unknown
yes
default
0
5000
10000
15000
20000
25000
30000
count

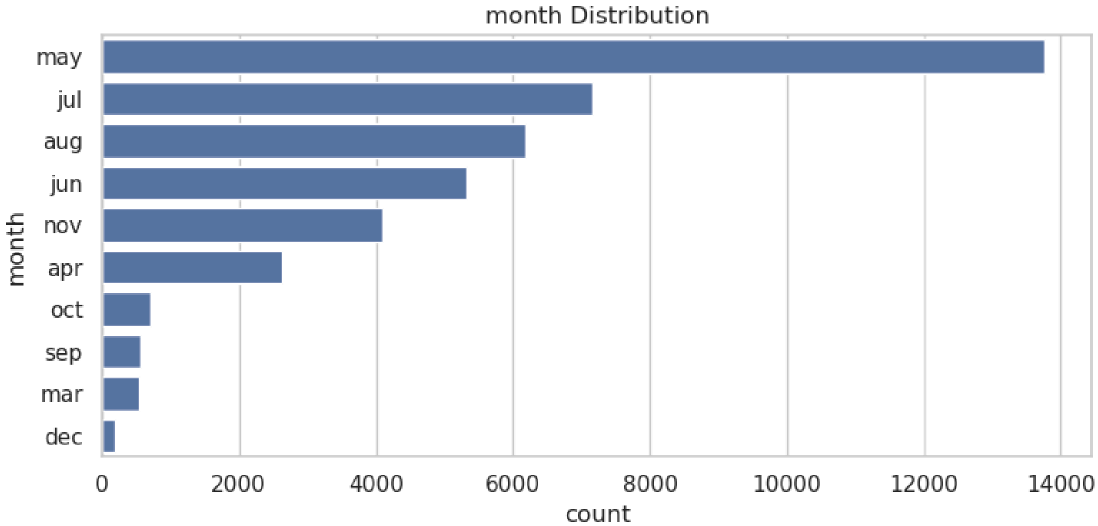

education Distribution
university.degree
high.school
basic.9y
professional.course
basic.4y
basic.6y
unknown
illiterate
education
0
2000
4000
6000
8000
10000
12000
count

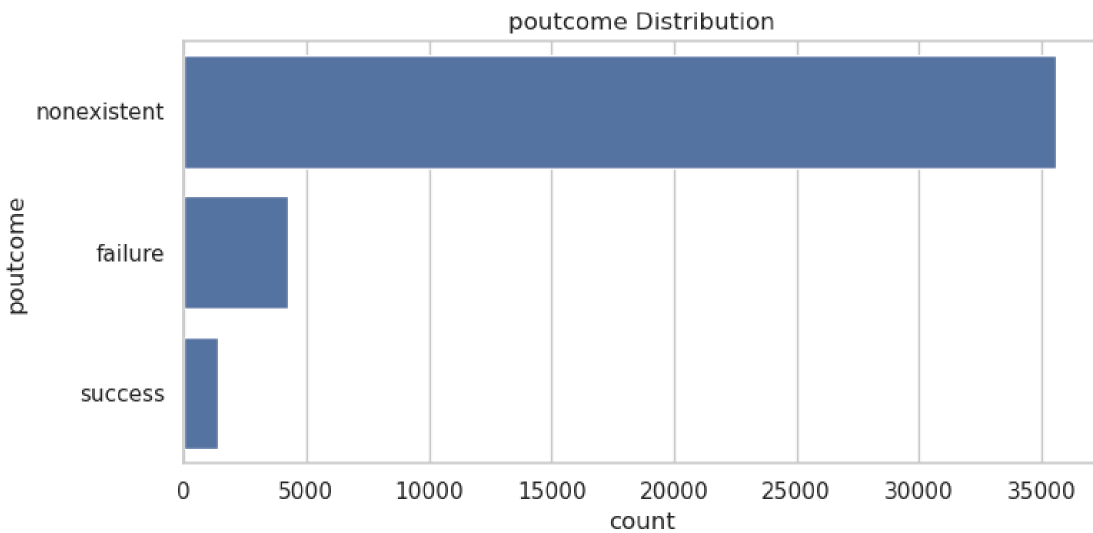

housing Distribution
housing
yes
no
unknown
0
5000
10000
15000
20000
count


job Distribution
job
admin.
blue-collar
technician
services
management
retired
entrepreneur
self-employed
housemaid
unemployed
student
unknown
0
2000
4000
6000
8000
10000
count


loan Distribution
no
yes
unknown
loan
0
5000
10000
15000
20000
25000
30000
35000
count

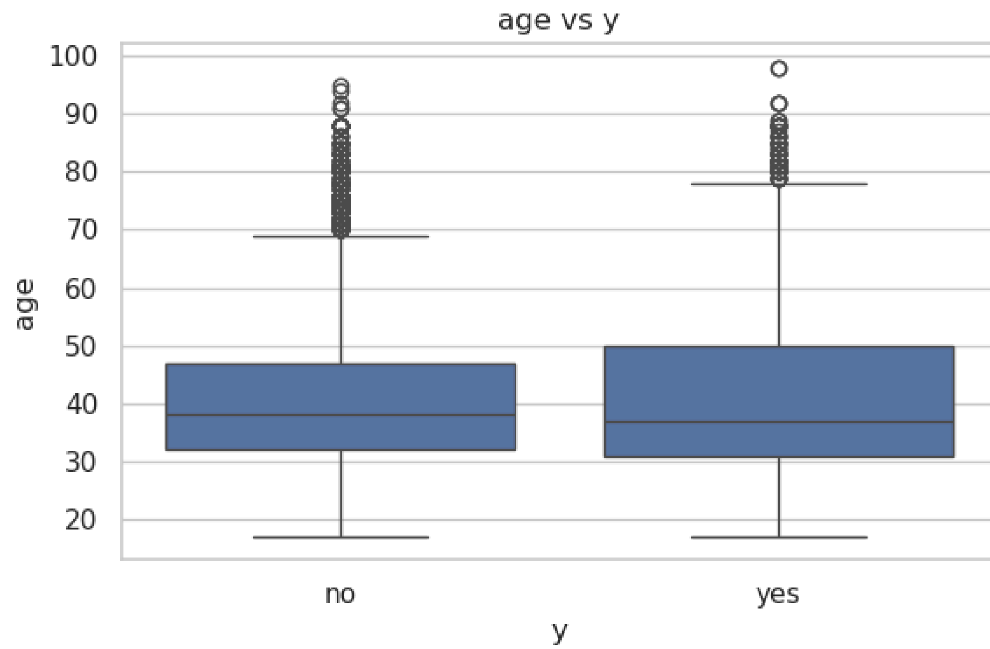

marital Distribution
married
single
divorced
unknown
marital
0
5000
10000
15000
20000
25000
count

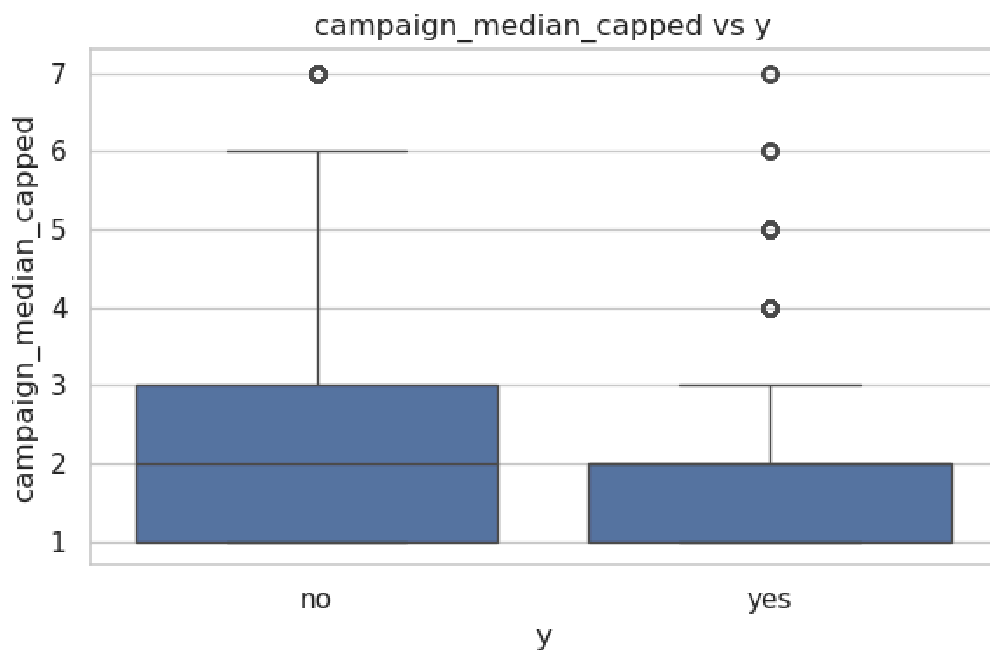

month Distribution
may
jul
aug
jun
nov
apr
oct
sep
mar
dec
month
0
2000
4000
6000
8000
10000
12000
14000
count

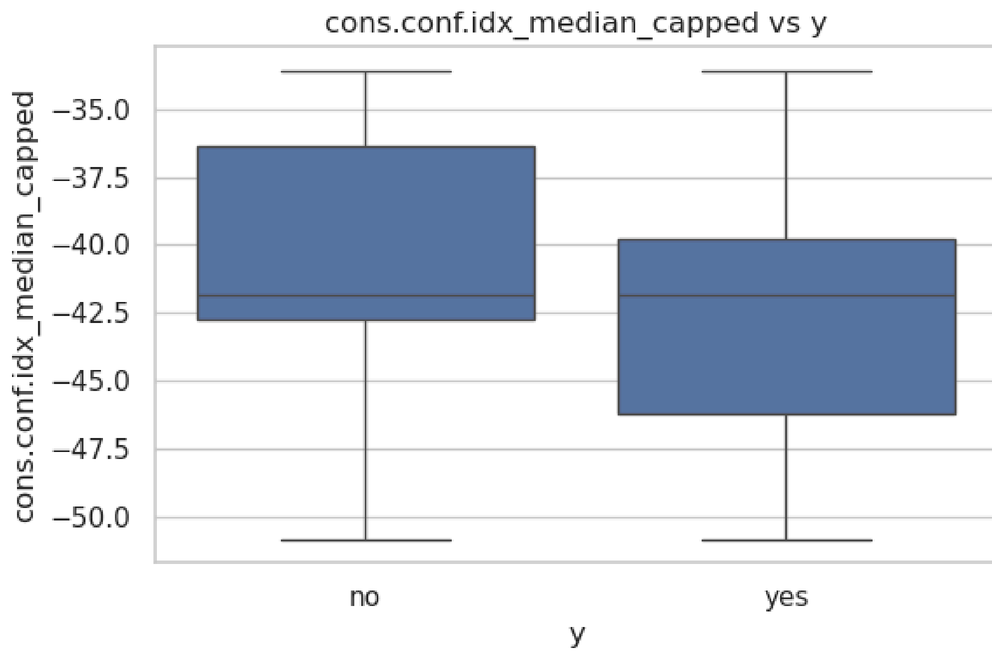

poutcome Distribution
nonexistent
failure
success
poutcome
0
5000
10000
15000
20000
25000
30000
35000
count

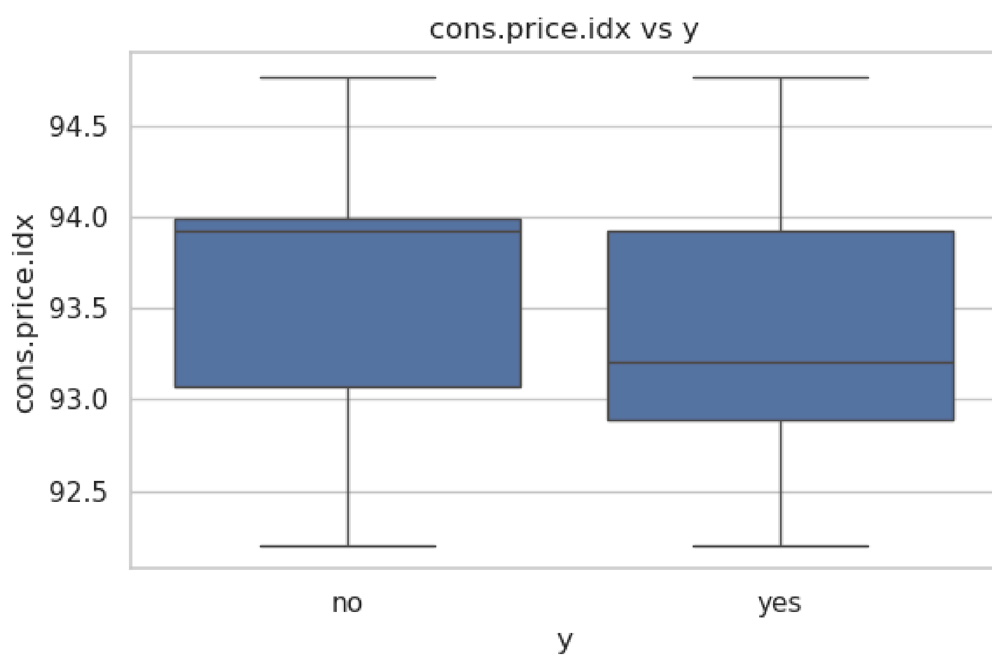
- **Box plot**

What to Look For	Interpretation
Clear separation in medians	Feature likely useful for prediction
Almost identical boxes	Feature likely **not** useful (little difference between yes/no)
Outliers or long tails	Investigate potential extreme values or skewed distributions
Wider spread in one class	Could suggest more variability in that class (e.g. behavior varies for buyers)

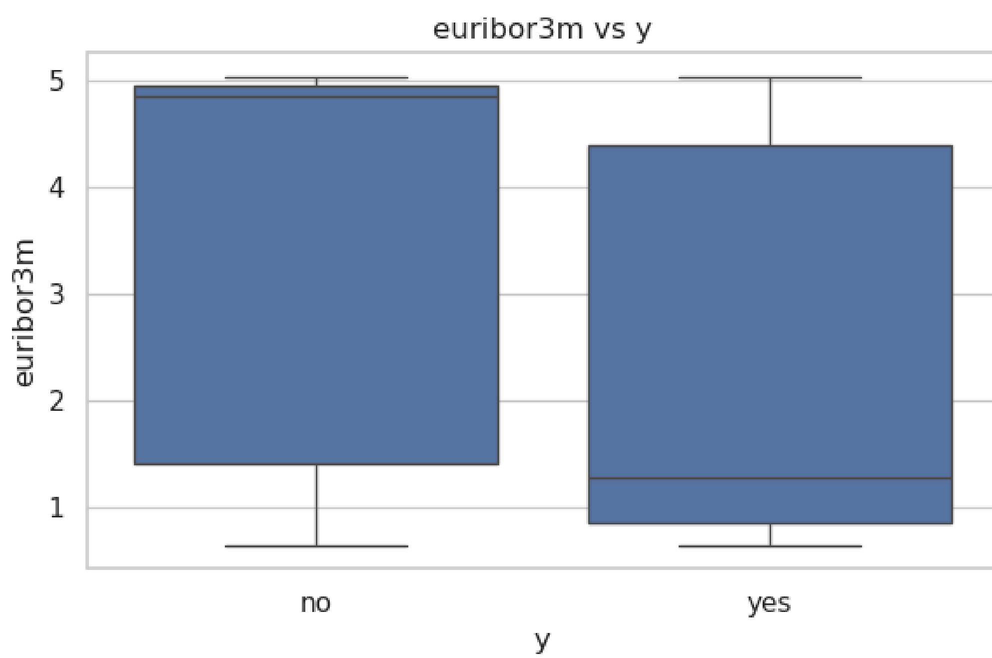

age vs y
100
90
80
70
60
50
40
30
20
age
no
yes
y

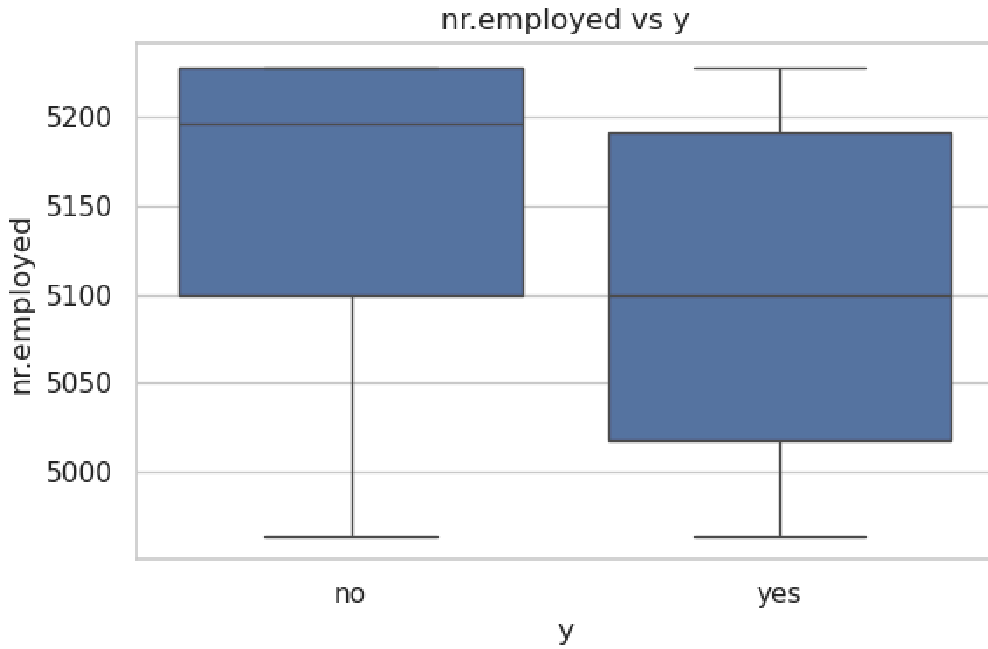

campaign_median_capped vs y
7
6
5
4
3
2
1
campaign_median_capped
no
yes
y

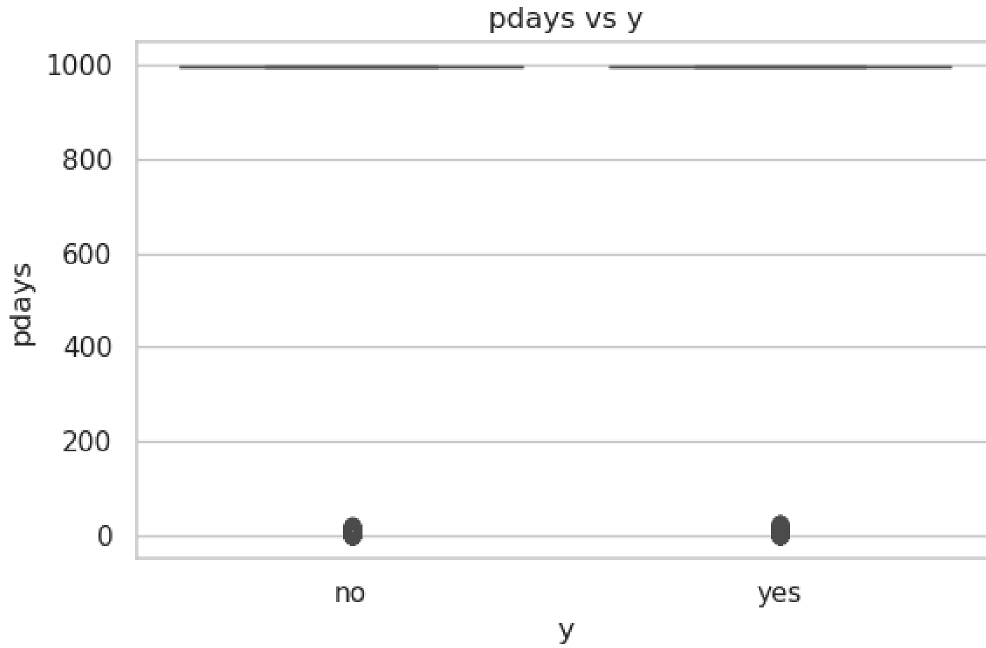

cons.conf.idx_median_capped vs y
cons.conf.idx_median_capped
−35.0
−37.5
−40.0
−42.5
−45.0
−47.5
−50.0
no
yes
y

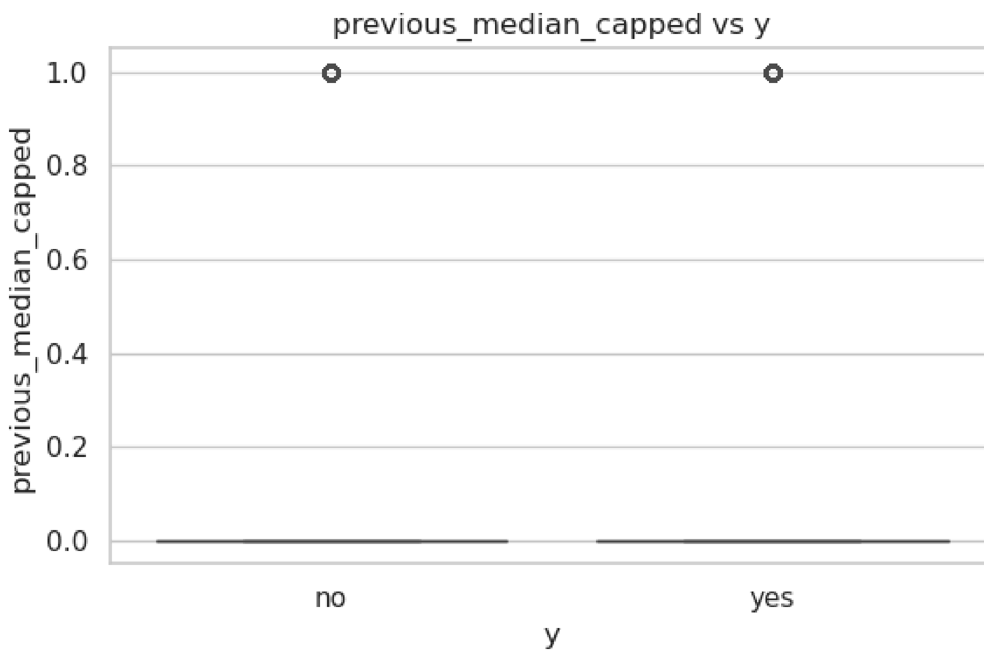

cons.price.idx vs y
cons.price.idx
94.5
94.0
93.5
93.0
92.5
no
yes
y

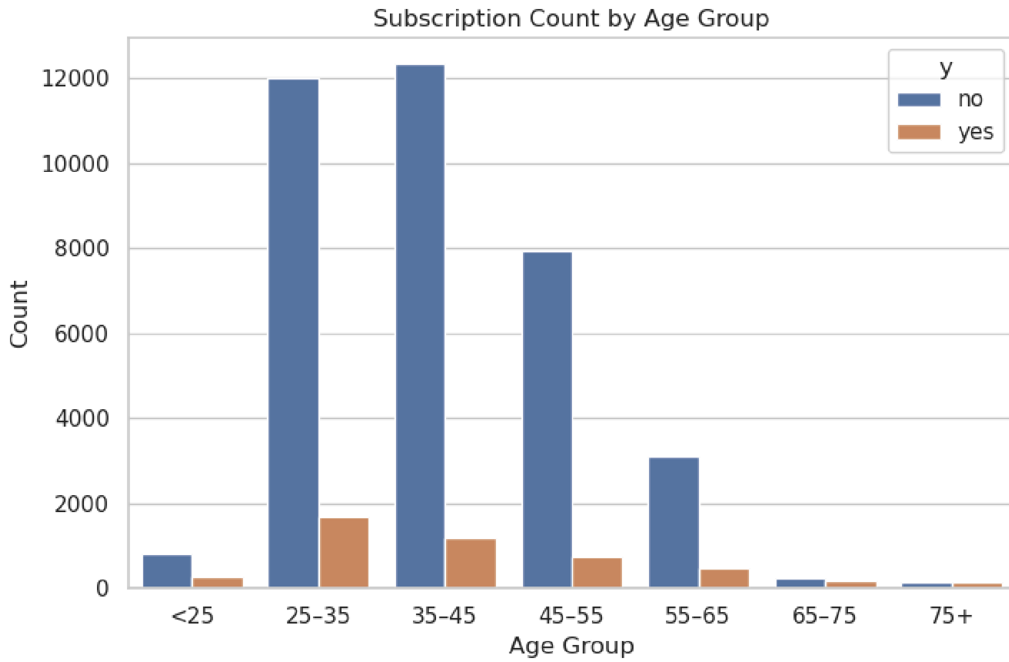

emp.var.rate vs y
emp.var.rate
1
0
−1
−2
−3
no
yes
y


euribor3m vs y
euribor3m
5
4
3
2
1
no
yes
y


nr.employed vs y
nr.employed
5200
5150
5100
5050
5000
no
yes
y


pdays vs y
1000
800
600
400
200
0
pdays
no
yes
y


previous_median_capped vs y
1.0
0.8
0.6
0.4
0.2
0.0
previous_median_capped
no
yes
y

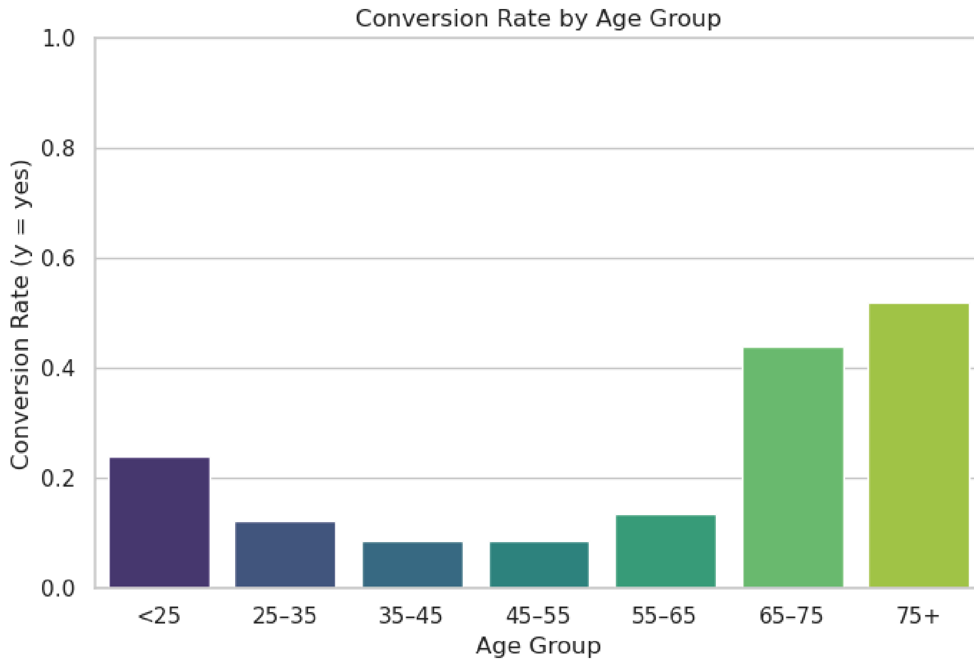


group_counts = df.groupby(['age_group', 'y']).size().unstack().fillna(0)

group_counts['count'] = group_counts['yes'] + group_counts['no']

group_counts['conversion_rate'] = group_counts['yes'] / group_counts['count']

conversion_rate: It tells you which age groups are **more likely to Customer says `yes` to subscribing to the bank's term deposit offer.** → helps focus marketing efforts where the probability is higher.



Data Preprocessing Steps:

1. Removed redundant column (`age_group`) created for EDA only.

2. Applied **one-hot encoding** to categorical variables (with `drop_first=True` to avoid multicollinearity).

3. Converted target variable `y` into **binary (1 = subscribed, 0 = not subscribed)**.

4. Verified shape change from original to encoded dataset.

4. Github Link :

https://github.com/priyanjalipatel/Data_Glacier_Final_Project/blob/main/Notebook_CLEAN_Tranformation.ipynb